MedTrust AI

Reliability-Aware Clinical Decision Routing

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PROJECT OVERVIEW

MedTrust AI is a research framework designed to investigate whether predictive confidence alone is sufficient to determine whether a clinical AI prediction should be accepted.

The framework combines:

1. Predictive confidence
2. Explanation reliability

The combined signals are used to route predictions into:

ACCEPT
REVIEW
DEFER

DATASET AND TASK

Dataset:
MIMIC-IV Clinical Database Demo v2.2

Task:
ICU mortality prediction

Feature window:
First 6 hours after ICU admission

Prediction target:
Death during the same ICU stay

MODEL PIPELINE

Baseline models:

- Logistic Regression
- Random Forest

Primary model:

- XGBoost

Explanation method:

- SHAP TreeExplainer

Explanation reliability prototype:

- SHAP stability under small controlled perturbations

FINAL DEMO RESULTS

Number of test patients:
21

Accuracy:
0.9048

Precision:
0.0000

Recall:
0.0000

F1:
0.0000

AUROC:
0.4211

Sensitivity:
0.0000

Specificity:
1.0000

Brier Score:
0.0887

RELIABILITY RESULTS

Average predictive confidence:
0.8814

Average explanation reliability:
0.9002

Average joint reliability:
0.8908

ROUTING RESULTS

ACCEPT:
20.0

REVIEW:
1.0

DEFER:
0.0

Acceptance coverage:
0.9524

Accepted error rate:
0.1000

INTERPRETATION

The prototype demonstrates how predictive confidence and explanation stability can be combined into a reliability-aware routing mechanism.

The current demonstration produced mostly ACCEPT decisions, with a smaller number of REVIEW decisions.

No DEFER decisions were produced under the current prototype thresholds.

IMPORTANT LIMITATIONS
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1. The demonstration uses the MIMIC-IV clinical database demo, which is very small compared with the full MIMIC-IV dataset.

2. Only a small number of ICU deaths are present in the demo.

3. Therefore the reported performance values are pipeline validation results and should NOT be interpreted as clinical performance claims.

4. The current explanation reliability score is a prototype based on SHAP stability.

5. A complete ERS should additionally investigate faithfulness, ranking consistency, and sign consistency.

6. The current ACCEPT/REVIEW/DEFER thresholds are prototype thresholds.

7. Final research thresholds should be selected using validation data and frozen before test evaluation.

8. Final research should use patient-level splitting, leakage-safe preprocessing, temporal validation, and external validation on eICU-CRD.

RESEARCH CONTRIBUTION

The central research contribution being investigated is the use of explanation reliability as an additional signal beyond predictive confidence for clinical decision routing.

The intended routing policy is:

High confidence + reliable explanation
-> ACCEPT

Uncertain or conflicting reliability signals
-> REVIEW

Low overall reliability
-> DEFER

FUTURE WORK

1. Run the pipeline on full MIMIC-IV.

2. Freeze the exact cohort and outcome definition.

3. Use patient-level and temporal splits.

4. Fit preprocessing only on training data.

5. Improve the Explanation Reliability Score.

6. Add explanation faithfulness.

7. Add feature ranking consistency.

8. Add sign consistency.

9. Select routing thresholds using validation data.

10. Evaluate selective risk.

11. Test temporal robustness.

12. Perform external validation on eICU-CRD.

13. Compare confidence-only routing against confidence + explanation reliability routing.

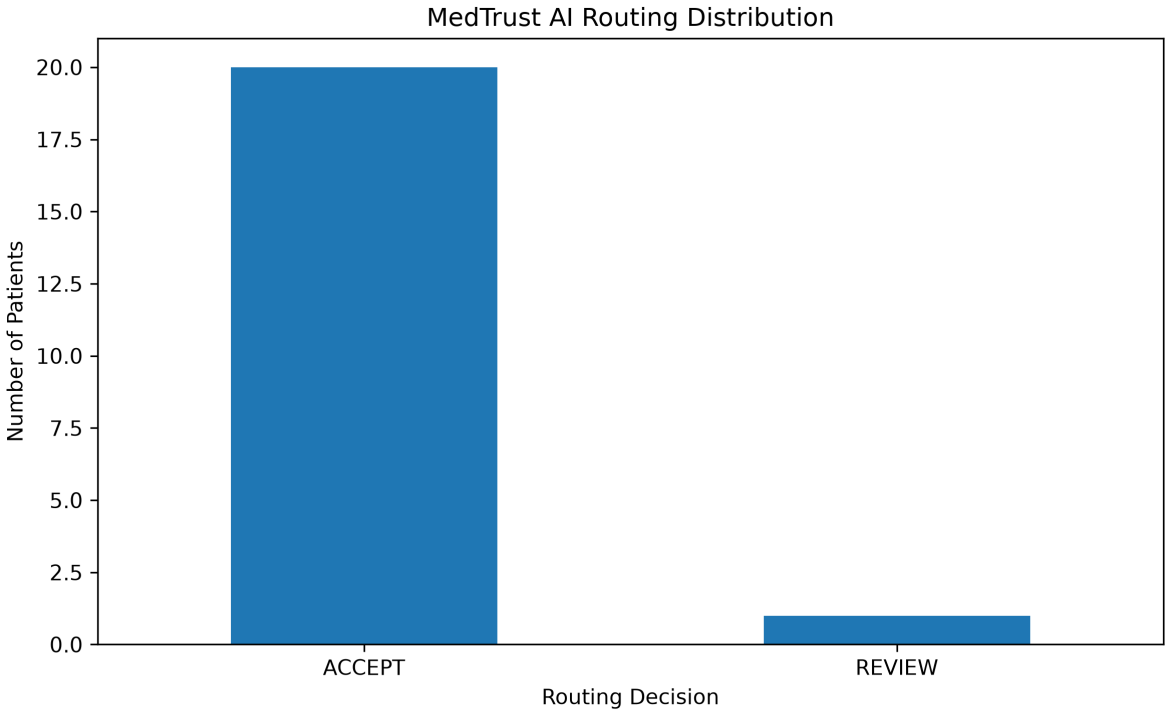
CONCLUSION

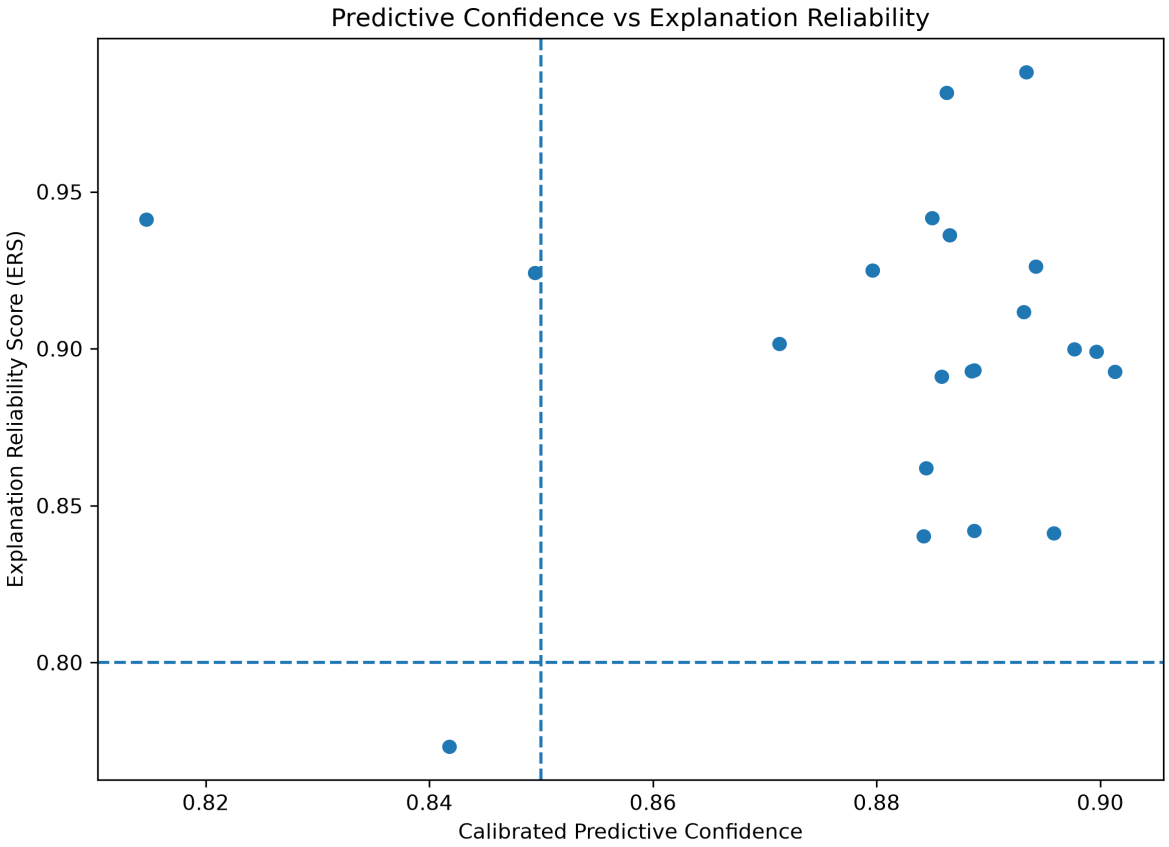
The current prototype successfully implements the main MedTrust AI pipeline from prediction to explanation and reliability-aware routing.

The demo establishes the technical workflow but does not establish clinical effectiveness. Larger-scale validation is required before making clinical or general performance claims.

Final Metrics

Metric	Value
Accuracy	0.9048
Precision	0.0000
Recall	0.0000
F1 Score	0.0000
AUROC	0.4211
Sensitivity	0.0000
Specificity	1.0000
Brier Score	0.0887
Average Predictive Confidence	0.8814
Average Explanation Reliability	0.9002
Average Joint Reliability	0.8908
ACCEPT Count	20.0000
REVIEW Count	1.0000
DEFER Count	0.0000
Acceptance Coverage	0.9524
Accepted Error Rate	0.1000


MedTrust AI Routing Distribution
Number of Patients
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
ACCEPT
REVIEW
Routing Decision


Predictive Confidence vs Explanation Reliability
Explanation Reliability Score (ERS)
0.95
0.90
0.85
0.80
0.82
0.84
0.86
0.88
0.90
Calibrated Predictive Confidence